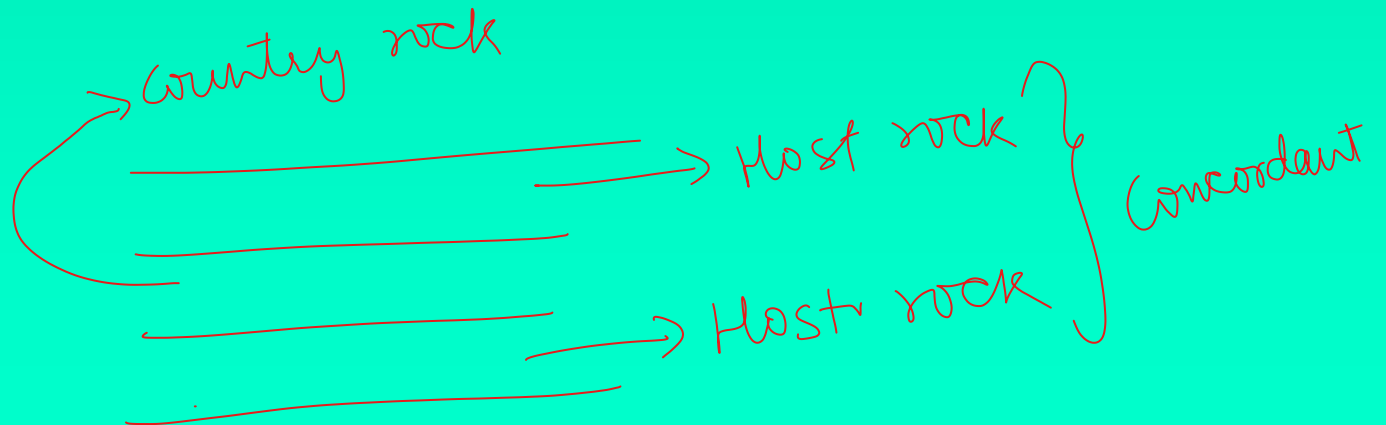


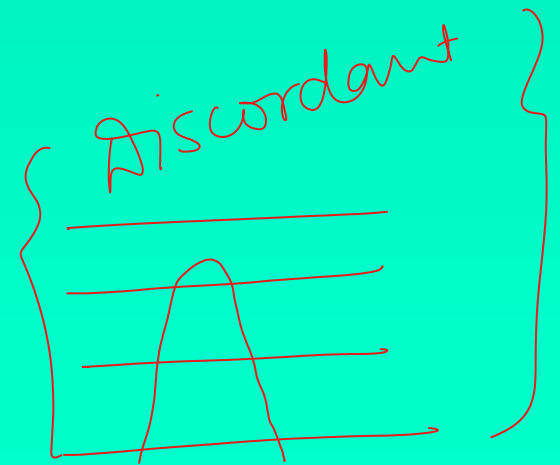
Q. what are intrusive rock?

INTRUSIVE IGNEOUS ROCKS



- Based on the attitudes of the associated country rocks the forms are called either as **Concordant or Discordant**.
- Concordant plutons runs parallel to the bedding or schistosity of the adjacent rocks.
- Discordant plutons cut across the structure of older formation.

Concordant → parallel
Discordant → traverse



FORMS OF INTRUSIVE IGNEOUS ROCKS

- Commonly observed forms of Plutonic (intrusive) rocks observed in the field are: **dykes, sills, laccoliths, dykes, phacoliths, lopolith, volcanic necks, batholiths and chonoliths.**

→ Dykes

→ sills

→ laccoliths

→ Dykes

→ phacoliths

→ lopolith

→ volcanic necks

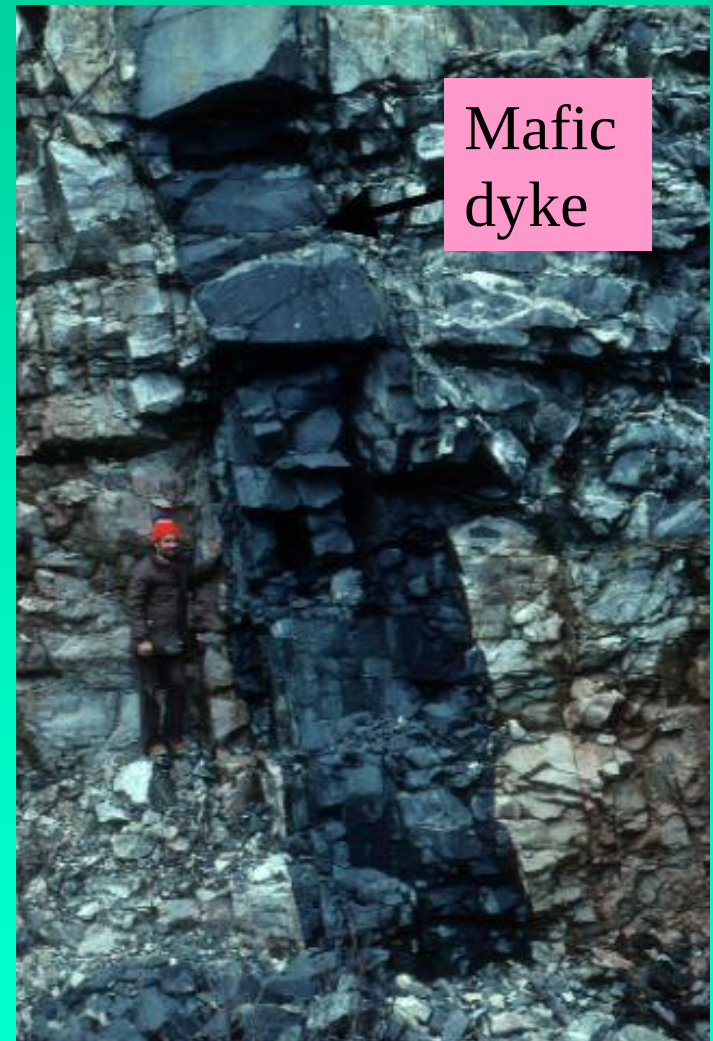
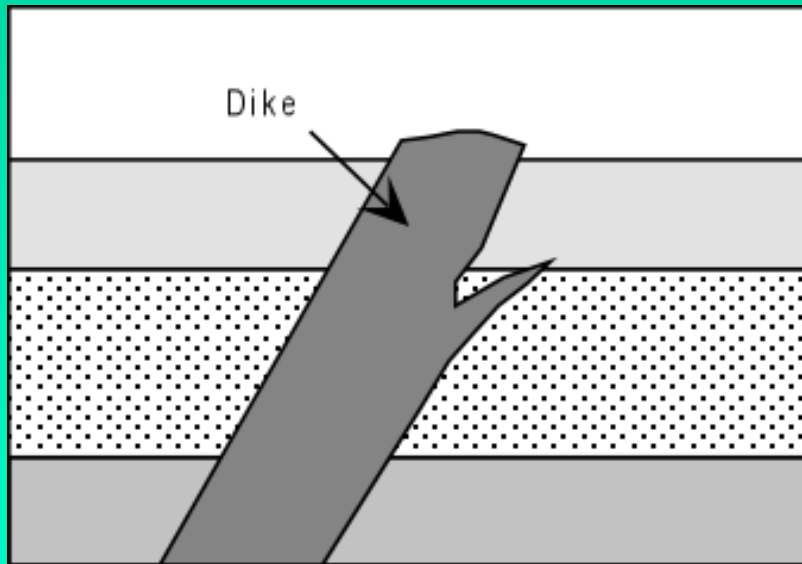
→ Batholiths

→ chonoliths

DYKES: Dykes and sills are the most common forms of the intrusive igneous bodies.

- **They are discordant**
- **Cut across the bedding of the rocks in which they intrude**
- **Vertical to steeply inclined and sheetlike body (extensive in lateral dimension)**
- **Thickness vary widely from an inch upto hundred of feet**
- **Injected through fractures, joints, and weak planes**

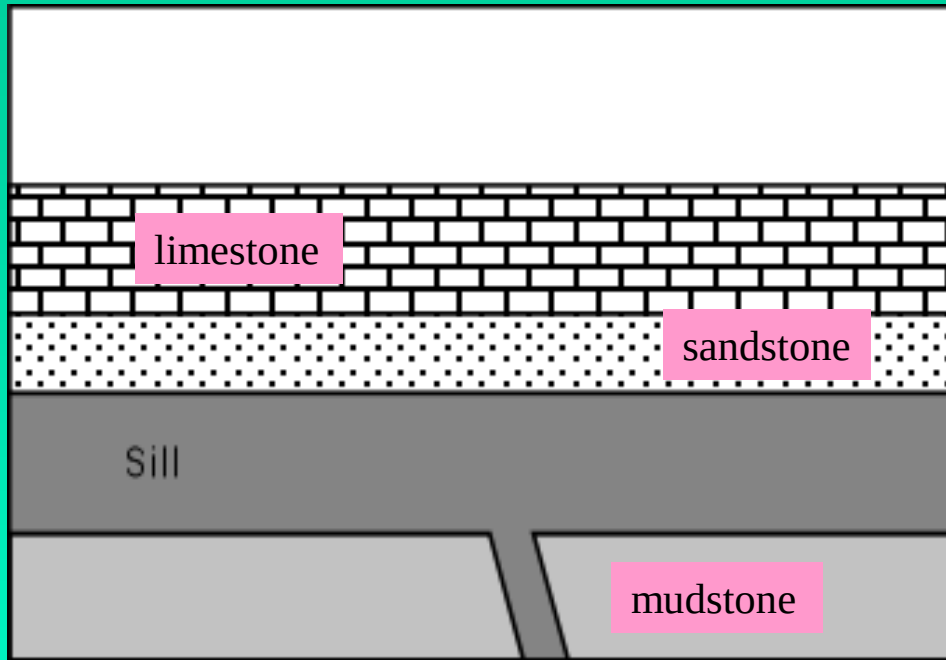
- Quartz-Dolerite dykes of Midland valley of Scotland are about 50-60 km long and upto 30m thick. Few places some dykes are very short upto few meters and as thin as few cm.



eg. Lamprophyric rocks broadly represented by *kimberlites*.
Lamprophyre dykes are located close to the western and eastern margins of the Cuddapah basin in the eastern Dharwar craton of India.

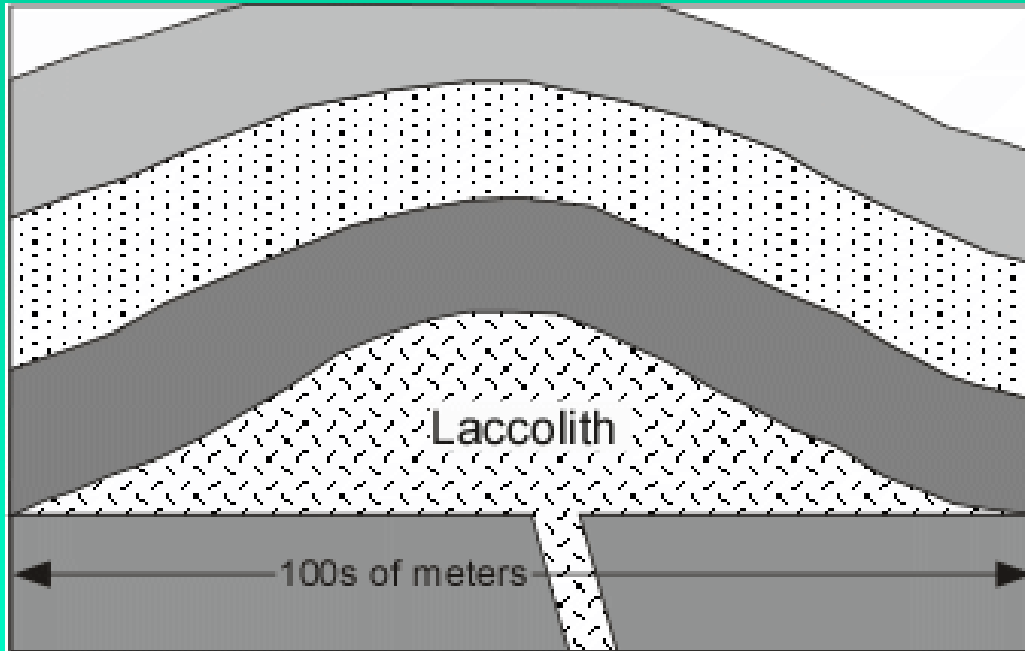
SILLS: Sills are relatively thin tabular sheetlike body that penetrates parallel to the bedding planes.

- Sills: Sills are relatively thin tabular sheet-like body that penetrates parallel to the bedding planes.
- They may be horizontal vertical or inclined. They are younger than rocks on either side of it.
- Laterally it may extends for 100s of km and upto 10 km in width.
- Lateral extend mainly depends on the hydrostatic force, temperature, degree of fluidity or viscosity, weight of overlying sediment column.
- Since basic magma are more fluid then acidic magma- mostly sills are made up of gabbros, dolerites and basalts.



Spreads parallel to the bedding planes of the rocks, hence concordant in nature.

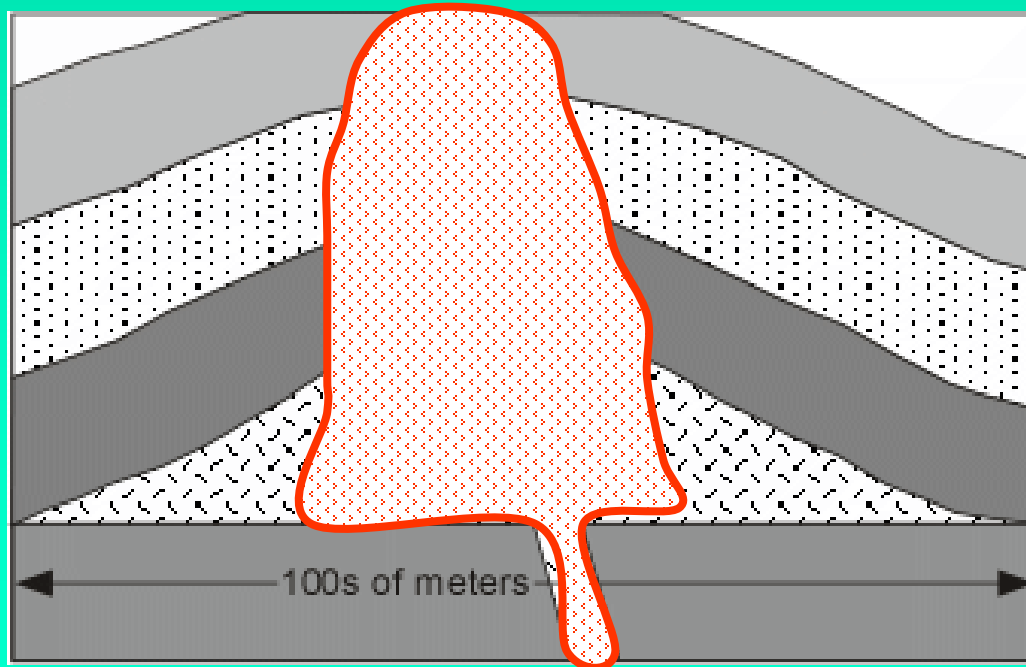
LACCOLITHS: It is a concordant body, with flat bottom and convex upward. It is dome shaped.



- When viscous magma is injected rapidly along the bedding, as it cannot spread it pushes up the overlying layers and keep on piling up.
- It causes folding of the overlying rock layers.

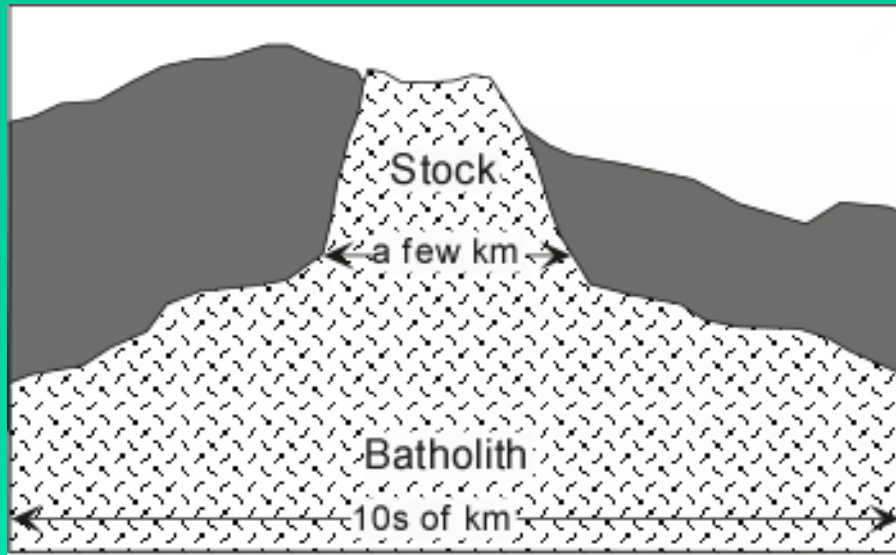
BYSMALITH: It is cylindrically shaped body of which the roof was uplifted along a circular or arcuate fault.

- It is developed when highly viscous magma is injected, because the lateral spreading along the bedding is less it acquires to move upwards and form cylindrical shape.
- Causes breaking of overlying rock layers.

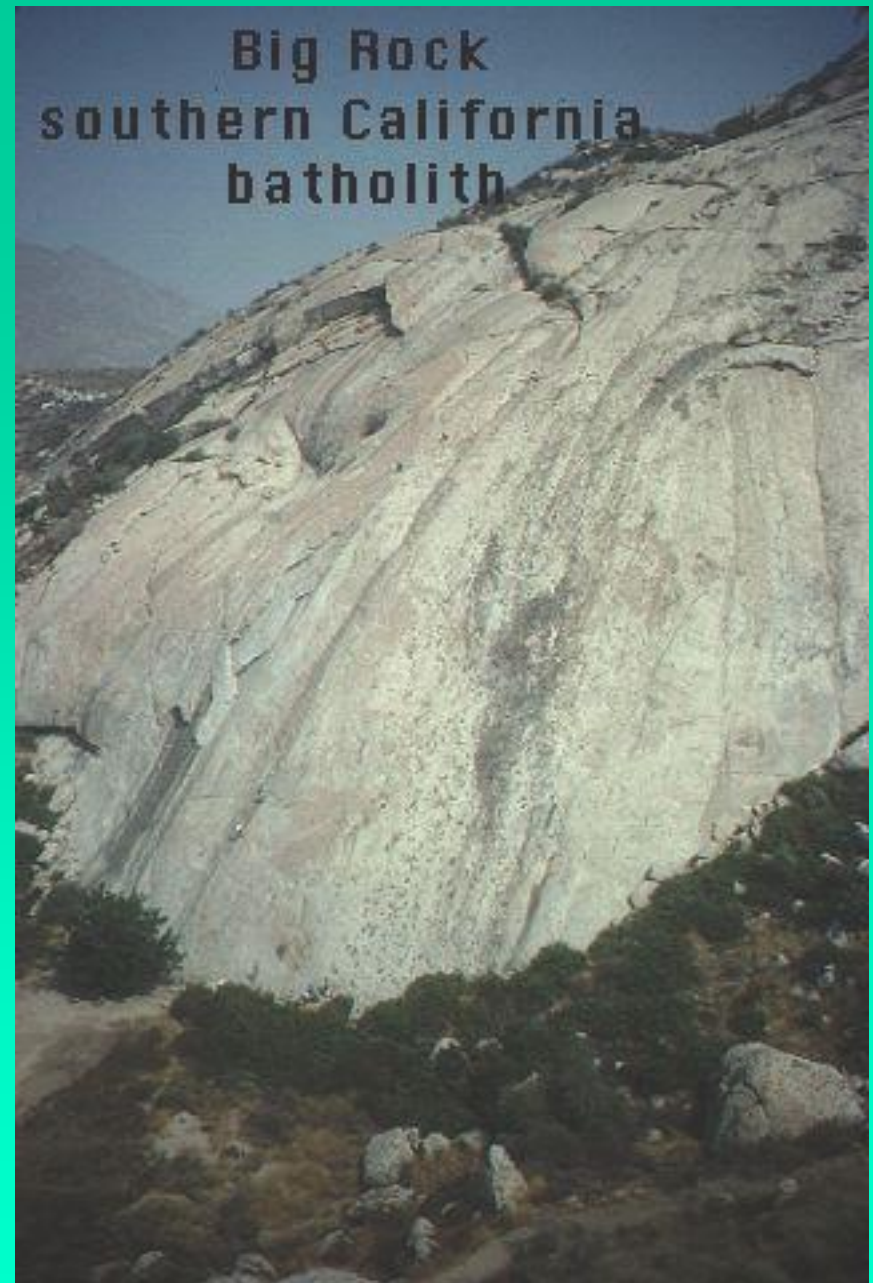


BATHOLITHS: are the largest kind of plutons, irregular in shape and occupies large area.

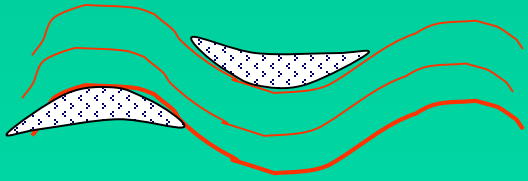
- Their side sloping away from each other which makes them larger and large downwards extending to greater depth
- Their occurrence is commonly associated with the mountain-building process
- These are either granites or granodiorites in composition



Stocks: *Are smaller irregular bodies with 10 km in maximum dimension, and are associated with batholiths.*

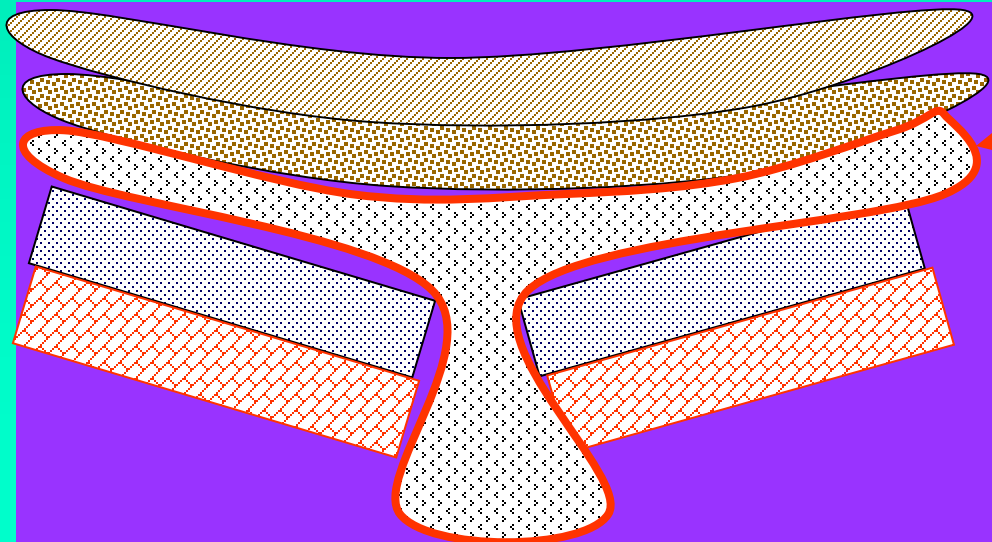


PHACOLITHS: These are concordant bodies that occurs along the crests and troughs of the folded sedimentary strata.



LOPOLITHS: These are basin or saucer-shaped concordant bodies with top nearly flat and convex bottom

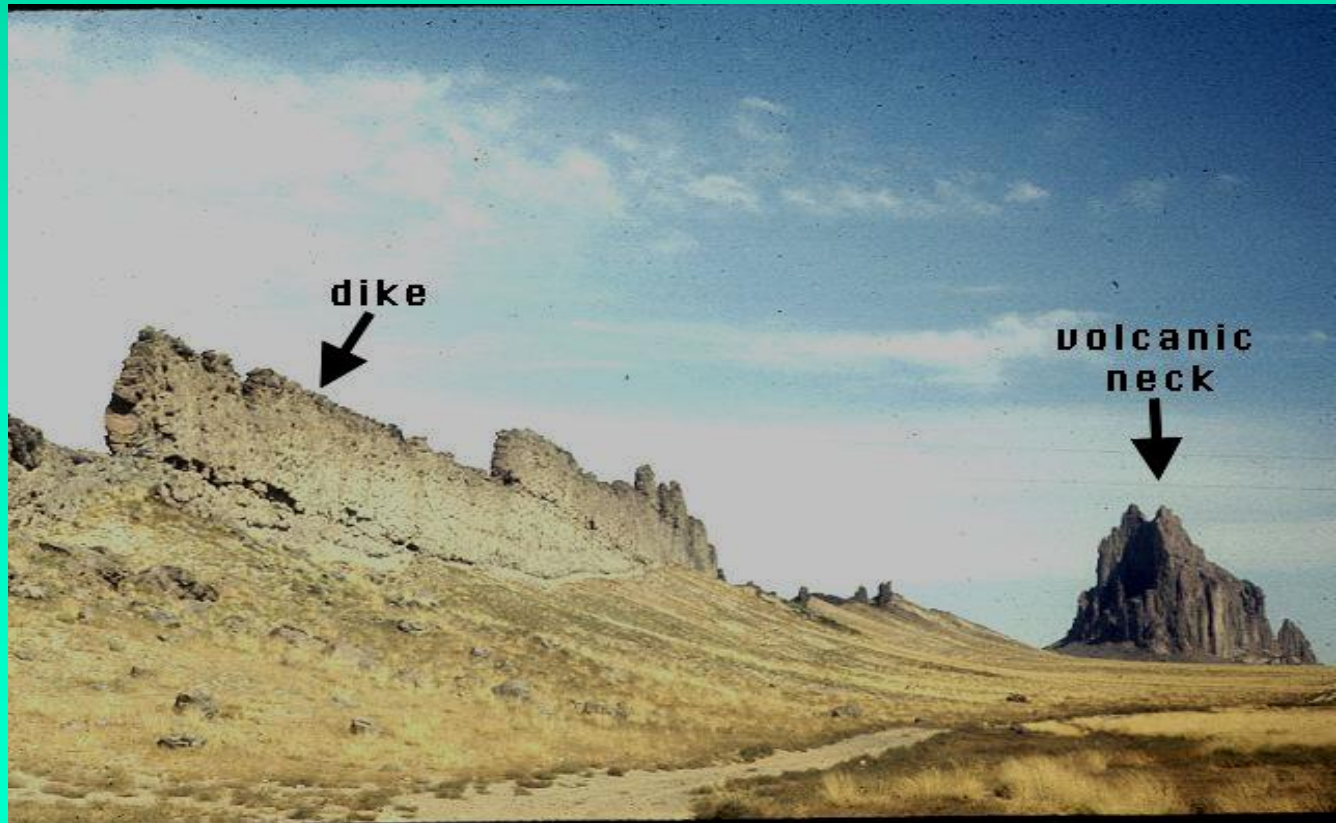
- They are very huge body with diameter upto 150 miles (app. 240 km)



Lopolith

CHONOLITHS: This term is applied to all other intrusive igneous bodies with irregular shape, i.e. the body with no specific shape.

VOLCANIC NECK or VOLCANIC PLUGS: It is cylindrical conduit that fed magma upward to a volcanic vent or it is a conduit of the ancient volcano. Vary in diameter from a few 100s of m to a kilometer or more. These are filled up with crystalline rocks. Shape-circular, elliptical or irregular.



TEXTURES OF IGNEOUS ROCKS

- *Based on granularity or grain size:*

- grain size depends on physical conditions that prevailed during the time of Crystallization of magma

i.e. presence of volatiles, rate of cooling, pressure, temperature etc.

Slow cooling = larger mineral grains

Fast cooling = smaller mineral grains

PHANERIC TEXTURE

Is characterized by **LARGE SIZE MINERALS** which can be easily seen by Naked eye (size at least 2mm or greater)

Coarse-grained
Phaneric

- > 5mm

Medium-grained
Phaneric

- 1 mm - 5mm

Fine-grained
Phaneric

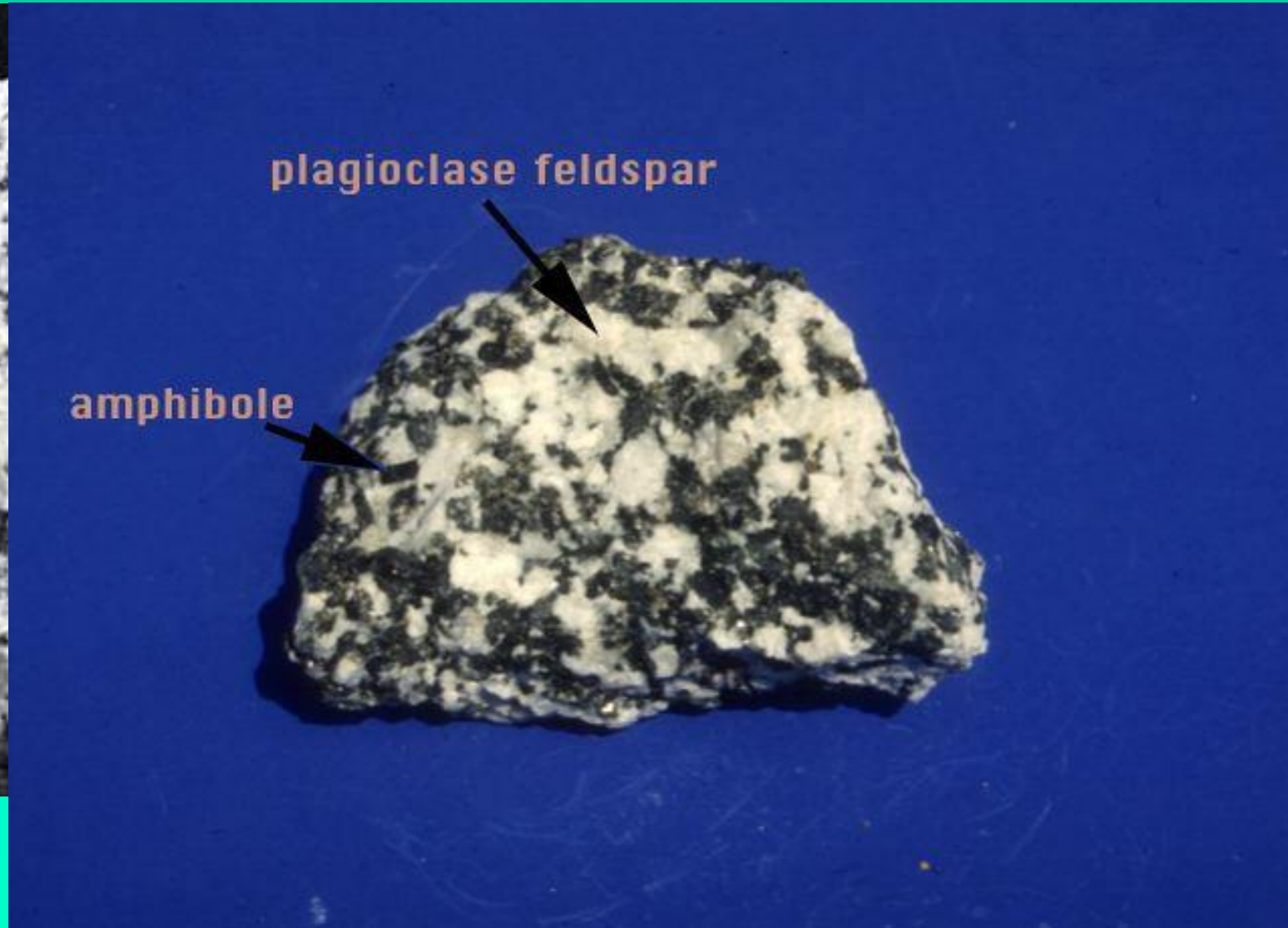
<1 mm

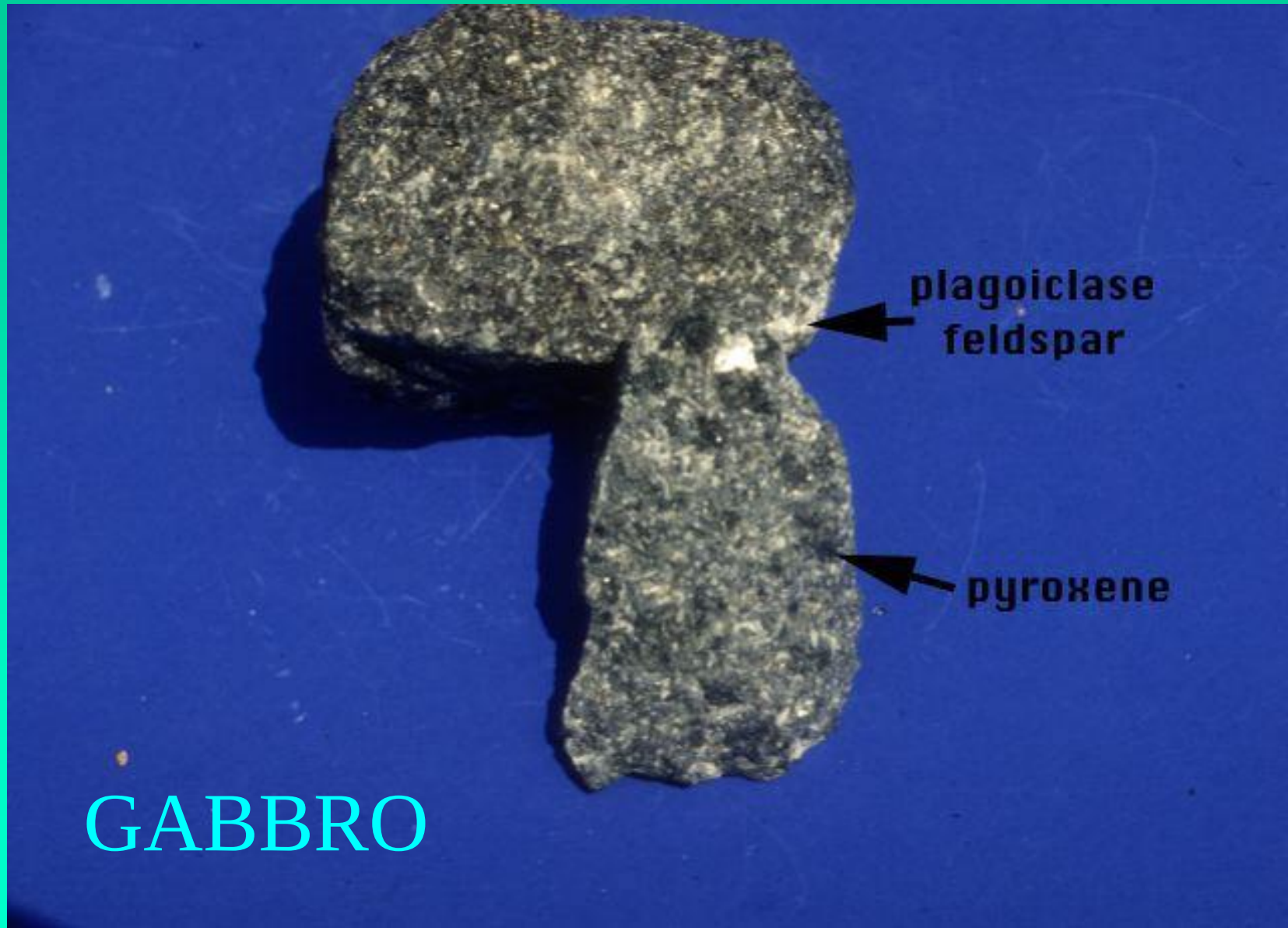
Commonly associated with the INTRUSIVE (PLUTONIC) IGNEOUS ROCKS, because magma in the crust cools at slower rate and have enough time to result into large mineral grains.

Eg. Granite; Pegmatite



DIORITE





GABBRO

APHANITIC TEXTURE

Is characterized by FINE GRAINED MINERALS, which can be seen under microscope (size < 2mm)

Commonly associated with the VOLCANIC (EXTRUSIVE) IGNEOUS ROCKS, because magma on the surface flows cools faster.

Eg. Basalts; Rhyolite



Basalt



Rhyolite

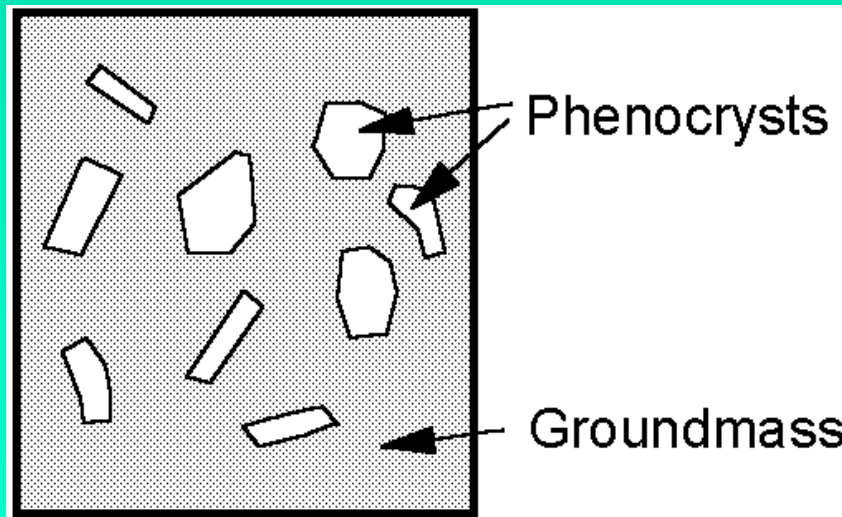
Obsidian



PORPHYRITIC TEXTURE

Is a distinctive mixture of large and fine grained mineral grains

This happens when slow cooling is followed by rapid cooling.
Phenocrysts = larger crystals, **matrix** or **groundmass** = smaller crystals



ANDESITE PROPHERY



Phenocrysts of Amphibole

Ground mass mainly of
Plagioclase feldspar (Na/Ca)